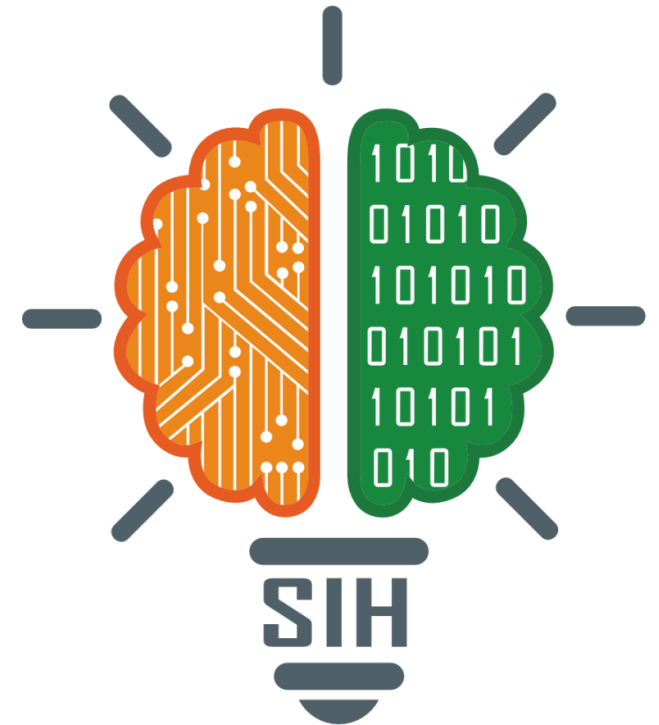


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26034**
- **Problem Statement Title - Software System to check compliance of Packaged Commodities under Legal Metrology**
- **Theme- Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name - LabelSentinals**





Capture

AI / OCR
ExtractionRule
ValidationViolation
DetectionInspector
VerificationAutomated
Report

From Package Image to Evidence-Backed Legal Metrology Compliance

DETAILED
SOLUTION

- **AI-POWERED CAPTURE:** Uses smartphone photos/video with Computer Vision + OCR to read key package declarations.
- **STRUCTURED EXTRACTION:** Converts MRP, net quantity, dates, manufacturer/packer/importer and other fields into structured data.
- **INSPECTION WORKFLOW:** Links every finding to the inspection case and captured evidence.

HOW IT
ADDRESSES
THE PROBLEM

- **FASTER INSPECTION:** Reduces repetitive manual reading and checking of package labels.
- **RULE-BASED VALIDATION:** A category-aware Legal Metrology Rule Engine checks applicable requirements.
- **EVIDENCE + VERIFICATION:** Highlights the relevant image region; confidence scores let inspectors approve or override findings.

INNOVATION &
UNIQUENESS

- **NOT JUST OCR:** Combines Computer Vision, OCR, information extraction and legal-rule validation.
- **EVIDENCE-BACKED FINDINGS:** Every potential violation is linked to visual evidence for review.
- **HUMAN-IN-THE-LOOP:** Inspector remains the final decision-maker before report generation.

OUTPUT: COMPLIANCE STATUS • VIOLATION EVIDENCE • PDF / EDITABLE INSPECTION REPORT

1. FRONTEND UI
(Progressive Web App)**React 19**
(TypeScript)**Vite****Tailwind CSS v4****Lucide React**
(Scalable Icons)

- Capture Label
- Crop & Preview
- View Results
- Generate Report

**Progressive
Web App (PWA)****2. BACKEND API
& LOGIC****FastAPI**
(Python)

- High-performance
- Asynchronous Endpoints
- RESTful API
- Authentication & Authorization

**3. DATABASE &
ORM****PostgreSQL****SQLAlchemy**

(ORM)

**Alembic**
(Migrations)

- Relational Database
- ACID Compliance
- Secure & Scalable

**4. AI & COMPUTER
VISION CORE****PaddleOCR**
(PaddlePaddle)Robust Text Extraction
from Complex Labels**Google
GenAI**
(Gemini)Legal Metrology
Rule Engine &
NLP Validation

Rule Engine



NLP Validation

**5. DOCUMENT
GENERATION****WEASYPRINT**
(HTML to PDF)**FPDF2**

(PDF Generation)



- Compliance Report
- Evidence Attachments
- Audit Ready
- Digital Signature Support

**IMPLEMENTATION PROCESS****1 IMAGE ACQUISITION &
PREPROCESSING**

The PWA captures the product label image. **react-easy-crop** allows the inspector to focus on relevant label areas before the backend processes it.

**2 OCR & TEXT EXTRACTION**

PaddleOCR analyzes the image to accurately extract raw text, handling different fonts, rotations, and complex packaging layouts.

**3 AI RULE ENGINE VALIDATION**

The raw text is passed to the Generative AI engine. The engine acts as a "Legal Metrology Expert," structuring the data and mapping it against **mandatory FSSAI/** Metrology rules.

**4 INSPECTOR VERIFICATION
(HUMAN-IN-THE-LOOP)**

The extracted data and compliance scores are sent back to the React UI. The inspector reviews the AI's findings, visually verifying the flagged evidence.

**5 FINALIZATION & REPORTING**

Once verified, the compliance status is securely saved to PostgreSQL, and a formal, evidence-backed PDF report is generated for auditing.

TECHNOLOGIES TO BE USED**React 19**
(TypeScript)**Vite****Vite****Tailwind
CSS v4****Lucide
React****Python****FastAPI****PostgreSQL****SQLAlchemy****Alembic****PaddleOCR****Google
GenAI****WeasyPrint****FPDF2**

Feasibility

- **Technically highly feasible** with existing AI technologies like PaddleOCR and GenAI.
- **Operationally simple** using standard smartphone cameras, ensuring immediate usability for inspectors and QA teams.
- **Economically viable** by automating manual inspection, reducing overhead and scaling cost-effectively on cloud.

Viability

- User friendly interface and **dashboards** drive both farmer and official engagement.
- Farm based **rewards** and community recognition boost participation in compliance and reporting.
- Active participation in **crop switching** initiatives can be incentivized with reward points.

Practical Implementation

- MVP ready for **pilot rollout in real farms** within **6 to 9 months**.
- Works **offline** for remote rural areas with seamless data sync.
- Simple **multilingual** app and UI for effortless access by all farmers.



Potential Challenges and Risks

- 01 Image Quality & Variations:**
Poor lighting, blurriness, or curved/reflective packaging can cause OCR failures.
- 02 Non-Standardized Layouts:**
Food labels have massive variations in design, making template-matching impossible.
- 03 OCR Accuracy Limitations:**
Complex fonts, clustered text, or local languages may lead to misread information.
- 04 Regulatory Changes:**
Frequent updates to Legal Metrology rules could make hardcoded compliance logic obsolete.
- 05 Connectivity Issues:**
Patchy internet in rural areas may disrupt real-time data sync and updates.

Pre-processing & Feedback:

Allow cropping with react-easy-crop and enhance images before OCR for better results.

AI vs Templates:

Use a Generative AI Rule Engine to parse unstructured text and identify data regardless of layout.

Human-in-the-Loop:

Show AI confidence scores and require final inspector verification to ensure accuracy.

Centralized Rule Engine:

Rules managed in backend (FastAPI/PostgreSQL) enabling instant updates to regulations.

Offline-First & Sync:

Enable offline capture and local storage with auto sync once internet is available.



Strategies For Overcoming Challenges

01
02
03
04
05

POTENTIAL IMPACT ON TARGET AUDIENCE

**Field Officers & Inspectors**

10× faster audits: 25–30 min → under 4 sec.
Automated OCR + AI checks 12 mandatory Rule 6 declarations.
Instant Form-1 memos, evidence photos & audit trails.

**Consumers & Citizens**

Protects MRP, quantity and consumer-care declarations.
Supports public verification, photo complaints and real-time docket tracking.

**Manufacturers & Retailers**

Creates a level playing field against deceptive packaging and duplication.
Pre-market screening against Legal Metrology & FSSAI requirements.

KEY SOLUTION BENEFITS

**Faster Enforcement**

Rapid label reading, automated checks and digital evidence reduce inspection effort.

**Transparency & Trust**

Signed officer verification, traceable records and evidence-backed decisions improve accountability.

**Multilingual Access**

Detects package other regional language aging text across English, Hindi, Telugu .

**Paperless Enforcement**

Digital dossiers, inspection histories and PDF reports reduce paperwork and storage.

ECONOMIC • SOCIAL • ENVIRONMENTAL

**Economic Benefits**

- Curbs illegal MRP markups & quantity deception.
- Speeds identification of non-compliance and penalties.
- Up to 90% lower administrative effort through digitisation.

**Social & Consumer Welfare**

- Improves confidence in price, quantity, dates and licences.
- Supports regional-language enforcement.
- AI cues + mandatory officer verification promote fair decisions.

**Environmental & Operational**

- Paperless inspection chain and cloud dossiers.
- Less physical evidence storage.
- Efficient image/OCR processing reduces unnecessary bandwidth and compute.

RESULT: Faster inspections • Stronger compliance • Better consumer protection • Digital, accountable enforcement

Statutory Acts, Regulations & Government Gazettes

[Legal Metrology Act, 2009 \(Act No. 1 of 2010\)](#)

Key Sections: Sec. 18 — Mandatory packaging declarations; Sec. 36 — Penalty for non-compliant packages; Sec. 48 — Compounding of offences.

[Legal Metrology \(Packaged Commodities\) Rules, 2011 & 2021 Amendment](#)

Rule 6(1) — Mandatory declarations; Rule 6(1)(k) — Unit Sale Price; Rule 9 — Font height & area ratio.

[FSSAI Labelling and Display Regulations, 2020](#)

FSSAI licence number, manufacture / best-before / expiry dates and allergen declarations.

AI, Computer Vision & Multimodal Research Papers

[PP-OCrv4 — Du et al. \(2023\)](#)

DBNet text detection + SVTR recognition for multilingual packaging text extraction.

[Gemini: A Family of Highly Capable Multimodal Models \(2023\)](#)

Cross-referencing detected text with statutory rules, dual-MRP detection and bounding-box localisation.

[Modified Laplacian — Pech-Pacheco et al. \(2000\)](#)

Variance of Laplacian sharpness filtering to reject motion blur and select sharp package panels.

Technical Implementation & Project Artifacts

[Live Deployed Application — Click Here to view](#)

[Platform Source Code \(GitHub\) — Click Here to view](#)

[National Open Packaging Corpus — Open Food Facts India Dataset](#)

[Retail Label Multi-Oriented Text Benchmark — Click Here to view](#)

Regulatory Compliance Context

[Consumer Protection \(E-Commerce\) Rules, 2020](#)

Supports digital marketplace compliance for pre-packaged commodities and consumer-facing product information.

[Ministry of Consumer Affairs, Food & Public Distribution](#)

[Food Safety and Standards Authority of India \(FSSAI\)](#)

LegalMetriX Research Application

1. 360° video capture → automatic keyframe extraction
2. Variance of Laplacian → sharpness filtering
3. PP-OCrv4 → multilingual text detection and recognition
4. Multimodal AI → entity extraction and statutory rule cross-checking
5. Compliance engine → evidence-backed violations and reporting

Quick Reference Links

[Legal Metrology Portal](#)

[FSSAI Official Website](#)

[PP-OCrv4 Research Paper \(arXiv\)](#)

[Gemini Research Paper \(arXiv\)](#)

[LegalMetriX Live Application](#)

[LegalMetriX GitHub Repository](#)